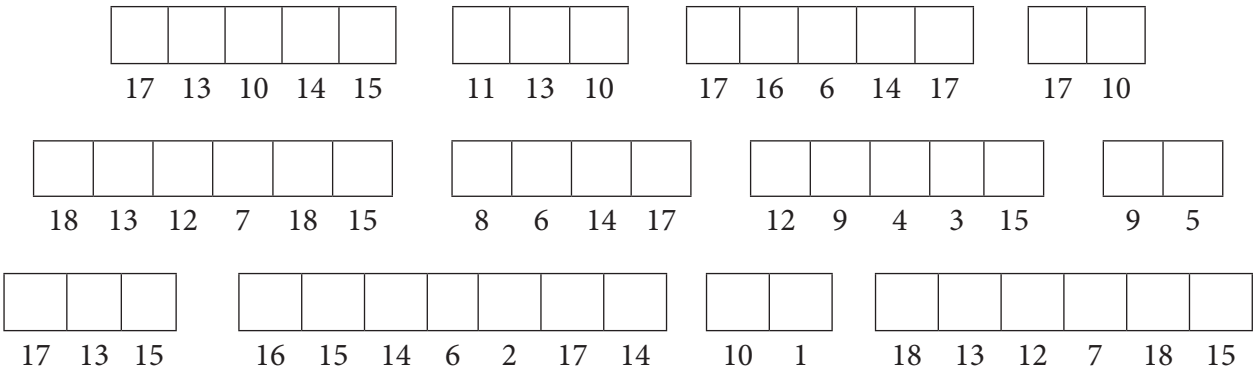
WORKSHEET

Combinations



John Calvin Coolidge

Questions

A softball team of 9 players and 2 reserves is to selected. In how many ways is this possible if:

1 15 people try out?

2 20 people try out?

3 12 people try out?

A class of 30 students work hard each week to be selected to receive donated movie tickets as a reward. In how many ways could the group to be rewarded be chosen if:

4 3 tickets are awarded?

5 5 tickets are awarded?

6 12 tickets are awarded?

Calculate the following:

7 ${}^{7}C_{4}$

8 ${}^{5}C_{3} + {}^{11}C_{8}$

9 ${}^{20}C_{19} \times {}^{4}C_{2}$

20 boys and 16 girls audition for the school play. In how many ways can the cast be chosen if there are:

10 4 male roles and 7 female roles?

11 12 male roles and 3 female roles?

12 5 male roles and 5 female roles?

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Zali and Coby are two of the eleven contenders for an elite swimming squad. In how many ways can the squad of 5 be chosen if:

13 Coby is in the squad but Zali isn't?

14 Both Zali and Coby are in the squad?

In a game of blackjack, a hand of 2 cards is dealt from a pack of 52 cards.

15 In how many ways can this be done?

16 How many hands have 2 red cards?

17 What is the probability that 2 black cards are dealt?

18 A winning hand of 21 involves being dealt an Ace and card of value 10 (i.e. 10, Jack, Queen, King). What is the probability of being dealt 21?

Solutions

A 67 721 472

B 120

C $\frac{32}{273}$

D 12

E 2652

F 1365

H 126

I 4060

L 167 960

M 175

N 35

O 55 426 800

R 650

S 84

T $\frac{25}{102}$

U 86 493 225

W 70 543 200

Y 142 506

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